

# Joint 3-class Lung Segmentation — SegResNet, pseudo-LIDC (full corpus w/ pseudo lung labels) — sliding-window ( $128^3$ ), CE weight 100

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## 1. Task

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End-to-end 3-class semantic segmentation of a full chest CT into {background, lung, nodule}. Same task as the earlier resize-recipe `joint_segresnet_pseudo` — the only difference is inference recipe: **sliding-window over  $128^3$  patches at native resolution** rather than a single forward on a  $256^3$  resample.

- **Input** (1, H, W, D) CT at native voxel grid.
- **Training input**  $128^3$  patches sampled via `RandCropByLabelClassesd` with class-balanced ratios `[bg, lung, nodule] = [0.2, 0.3, 0.5]`, `num_samples = 4` per volume.
- **Inference** MONAI `sliding_window_inference`, ROI (128, 128, 128), `overlap = 0.5`, `sw_batch_size = 4`, Gaussian blending, output at input resolution.
- **Output** (3, H, W, D) softmax logits: 0 = bg, 1 = lung, 2 = nodule.

## 2. Model

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| Property             | Value             |
|----------------------|-------------------|
| Architecture family  | SegResNet         |
| Spatial dims         | 3                 |
| Input channels       | 1                 |
| Output channels      | 3 (softmax)       |
| Init. feature width  | 16                |
| Trainable parameters | <b>20,663,555</b> |

## 3. Data

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| Source          | Role                                   |
|-----------------|----------------------------------------|
| NLST            | train + val + test                     |
| NSCLC-Radiomics | train + val + test                     |
| LIDC-IDRI       | excluded (no GT lung labels available) |

- **Split:** `unified.json` — patient-grouped, dataset-stratified.

- **Preprocessing:** none beyond intensity clipping / normalization; the full CT enters the sliding-window inference at its native voxel grid.

## 4. Training

| Setting         | Value                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Loss            | Multiclass Focal Tversky + weighted CE ( $\alpha=0.3$ , $\beta=0.7$ , $\gamma=2.0$ , $\lambda_{CE}=0.1$ , class weights [1, 1, 100]) |
| Optimizer       | Adam                                                                                                                                 |
| Learning rate   | $1 \times 10^{-5}$                                                                                                                   |
| Weight decay    | $1 \times 10^{-5}$                                                                                                                   |
| LR schedule     | Cosine annealing, $T_{max} = 400$ , $\eta_{min} = 1 \times 10^{-6}$                                                                  |
| Batch size      | 2 volumes $\times$ 4 patches = 8 patches / step                                                                                      |
| Patch size      | $128^3$ (class-biased random crop, ratios [bg, lung, nodule] = [0.2, 0.3, 0.5] )                                                     |
| Epochs          | 400                                                                                                                                  |
| Random seed     | 42                                                                                                                                   |
| Mixed precision | bf16                                                                                                                                 |
| Augmentation    | 3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast                                |
| Hardware        | 1 $\times$ NVIDIA H100 80 GB (fat01)                                                                                                 |

Best checkpoint recorded at **epoch 364 / 400**.

## 5. Evaluation protocol

Evaluated on the **unified test** split (325 series). All voxel-level metrics micro-averaged over the whole split. Per-class metrics computed against argmax predictions.

## 6. Results

### Ticket-required metrics (3-class average)

| Metric                                 | Value           |
|----------------------------------------|-----------------|
| <b>mean IoU (3-cls)</b>                | <b>0.7609</b>   |
| <b>mean IoU (lung + nodule)</b>        | <b>0.6440</b>   |
| <b>Accuracy</b>                        | <b>0.995203</b> |
| <b>Macro Precision (lung + nodule)</b> | <b>0.6638</b>   |

| Metric                       | Value  |
|------------------------------|--------|
| Macro Recall (lung + nodule) | 0.9235 |

### Per-class breakdown

| Class      | Dice (micro) | Recall | Precision | IoU    |
|------------|--------------|--------|-----------|--------|
| background | 0.9974       | 0.9971 | 0.9978    | 0.9949 |
| lung       | 0.9764       | 0.9768 | 0.9760    | 0.9539 |
| nodule     | 0.5008       | 0.8702 | 0.3516    | 0.3341 |

### Per-case Dice distribution — nodule class (325 test cases)

| Statistic | Value  |
|-----------|--------|
| Mean      | 0.3886 |
| Std       | 0.2518 |
| Min       | 0.0000 |
| p25       | 0.1703 |
| Median    | 0.3656 |
| p75       | 0.6015 |
| Max       | 0.8947 |

Voxels evaluated: 17,795,943,918. Cases: 325.

## 7. Notes on interpretation

Lung Dice at  $\approx 0.98$  is essentially saturated across all architectures and inference recipes — the lung is a large, well-defined foreground class. Nodule Dice is the real signal.

Sliding-window inference at  $128^3$  trades global lung context for preserved local resolution. On this corpus the trade tends to be neutral-to-negative on nodule Dice compared to the  $256^3$  resize baseline — patches see only  $\sim 15\%$  of a typical lung volume, which limits anatomy-conditional reasoning.

### Instance-level metrics (per-nodule)

Connected-component analysis with 26-connectivity, pooled over the held-out test split, on the native voxel grid.

- **Instance recall** — GT nodules touched by at least one predicted voxel, over all GT nodules.
- **Instance precision** — predicted components touching at least one GT voxel, over all predicted components.

| Metric                          | Value         |
|---------------------------------|---------------|
| GT nodules (components)         | 947           |
| GT nodules hit                  | 848           |
| <b>Instance recall</b>          | <b>0.8955</b> |
| Predicted components            | 11,044        |
| Predicted components hitting GT | 857           |
| <b>Instance precision</b>       | <b>0.0776</b> |

No minimum-size filtering is applied — every predicted component counts, so single-voxel false positives lower instance precision. Produced by `scripts/instance_metrics.py --config configs/joint_segresnet_pseudo_sw.yaml --ckpt <ckpt> --split test` (JSON: `results_instance/joint_segresnet_pseudo_sw.json`).

## 8. Reproducibility

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- Config `/home/werner/nodule-training-demo/configs/joint_segresnet_pseudo_sw.yaml`
- Checkpoint `stage3_joint/checkpoints_segresnet_pseudo_sw/best_model.pth`
- Metrics JSON `/tmp/claude-1002/-home-werner-nodule-segmentation/c2faa0a0-5a75-48f5-8288-377df40092b7/scratchpad/sw_metrics/joint_segresnet_pseudo_sw.json`
- Command `python stage3_joint/eval_metrics.py --config /home/werner/nodule-training-demo/configs/joint_segresnet_pseudo_sw.yaml --checkpoint stage3_joint/checkpoints_segresnet_pseudo_sw/best_model.pth --split test`